

MBI ENOW LEONARD APPELGRYN

Software Engineer | [linkedin.com/in/leonard-appelgryn](https://www.linkedin.com/in/leonard-appelgryn) | mbiapplegryn@gmail.com | Github

SKILLS

- **Languages:** JavaScript/TypeScript | SQL | HTML5 | CSS3
- **Tools & Frameworks:** React | NextJS | React-Native | Redux | Tailwind CSS | AWS | Zustand | Prisma | Drizzle ORM | React Query
- **Version Control:** Git

WORK EXPERIENCE

Global Bush Travel - Cameroon Douala

Nov 2025 - Present

Full Stack Engineer

- Engineered 8 end-to-end booking modules (flights, hotels, transfers, car rentals, airport lounge, e-SIM, insurance, visa) on a production Next.js 16 platform, scaling the agency's digital services to international markets.
- Architected a unified, secure payment system integrating MTN Mobile Money and Orange Money with server-side price verification across all modules, eliminating client-side price tampering.
- Designed a sessionless pre-payment booking pattern for 4 modules (e-SIM, insurance, visa, airport lounge), where booking records and prices are locked server-side before payment is processed.
- Implemented CSRF double-submit cookie protection, Upstash rate limiting, and reCAPTCHA v3 on sensitive endpoints, alongside a standardized payment error system with 14 error codes and EN/FR translations.
- Delivered full EN/FR internationalization (next-intl) across 80+ API routes and 50+ component libraries, with Zustand global state management and React Query for server-state caching.

TypeScript, Next.js, Tailwind CSS, Zustand, React Query, Prisma, Drizzle ORM, MySQL, Redis, Zod.

OneMarket - Cameroon Douala

Aug 2025 - Oct 2025

Mobile Dev Intern

- Implemented a user feedback/chat, thereby increasing user engagement by **20%**.
- Fixed UI and crash bugs, reducing crash reports by **10%**.
- Streamlined admin - user feedback loop, cutting average response time by **5%**.

JavaScript, React, React Native, Redux, Expo.

Tech Chantier - Cameroon Buea

Oct 2023 - Feb 2025

FrontEnd Engineer

- Enhanced and deployed a POS web app, boosting sales by **25%**, user satisfaction by **30%**, and reducing operational costs by **15%**.
- Optimized transactions by **50%** and cut bug reports by **60%** through rigorous testing.
- Executed a receipt module, reducing checkout time by **40%** and improving transaction accuracy by **25%**.
- Developed a digital platform connecting donors with those in need, increasing successful donations by **30%** and donor-recipient connections by **50%**.

JavaScript, TypeScript, React, Nextjs, React Native, Redux, Zustand, Expo, Tailwind CSS.

ProGuide - Cameroon Buea

May 2023 - Jan 2025

FrontEnd Lead

- Led frontend development, integrating gamified lessons and AI-driven recommendations, resulting in a **60%** increase in student engagement and a **35%** boost in course completion rates.
- Applied AI-powered insights for personalized assessments, leading to a **50%** reduction in grading time and a **30%** improvement in student performance.
- Developed a Progressive Web App (PWA) for offline access, cutting load times by **50%**, reducing bounce rates by **30%**, and increasing user retention by **45%**.

TypeScript, Nextjs, Tailwind CSS, Zod, Redux.

PROJECTS

Secure Image Management System

- Designed and implemented a secure authentication and authorization system for healthcare providers, including radiologists, ensuring controlled access to sensitive medical data.
- **Key Features:**
 1. **Role-Based Access Control:** Implemented a secure authentication and authorization system for healthcare providers, patients, and radiologists.

2. **Encrypted Storage & Sharing:** Enabled end-to-end encryption for image uploads and secure sharing with authorized personnel.
3. **Advanced Search & Categorization:** Integrated search by album name, file name, tags, categories, and creation date for efficient image retrieval.
4. **Favourites & Navigation:** Designed an intuitive image browsing experience, allowing users to filter, categorize, and navigate between favourite images seamlessly.

Flutter, TypeScript, Node.js, Express.js, MySQL, Prisma, Supabase.

EDUCATION

University of Buea - Cameroon Buea

B.Eng Computer Engineering